

Diagnosing the Evaluation Distribution: Argument-Taxonomy Distillation and an English Pivot for Climate Stance Detection in Hindi and Bengali

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Abstract

Public discussion of climate change on Indian video platforms is active, but labelled stance data in Indian languages is scarce. This paper describes our submission to the AISoMe 2026 shared task at FIRE 2026: given only English training resources, assign FAVOUR, AGAINST or NONE to Hindi and Bengali YouTube comments on the claim “*climate change and global warming is a serious concern*”. Before training any model, we examined the 1 000 evaluation comments and found strong evidence that they are machine-translated from English: they contain no code-mixing or emoji and refer to a United States science communicator rather than to India. This diagnosis motivated our system design. We built a 27-node taxonomy of climate arguments and used it to generate a balanced synthetic corpus, to supervise an auxiliary classification head, and to structure the prompts of a nine-model language-model panel (five teachers, three judges and one adversarial challenger), whose labels on the unlabelled evaluation set we distilled into compact encoders. The primary run translates the evaluation comments back into English and classifies them with an English-only encoder, which improves macro-F1 by about 0.10 over the best multilingual-encoder system, averaged over 300 paired half-splits of the development set. The submitted system reaches macro-F1 0.922 (Hindi) and 0.915 (Bengali) against held-out judge-panel labels (123 Hindi and 121 Bengali comments). Because these development labels are model-generated rather than human gold, we report their reliability evidence and discuss this limitation explicitly.

Keywords

stance detection, low-resource languages, Hindi, Bengali, knowledge distillation, argument mining, model calibration

1. Introduction

Stance detection asks how a text positions itself with respect to a fixed claim. It is not sentiment: a furious comment and an approving one can take the same stance, and a polite one can take none at all. In the AISoMe 2026 shared task at FIRE 2026, the text is a YouTube comment in Hindi or Bengali and the claim is “*climate change and global warming is a serious concern*”. The organisers supply English-only training resources and evaluate on the two Indian languages, so cross-lingual transfer is the nominal challenge. Runs are ranked by macro-F1, and more than eighty runs were submitted to the track.

Two properties of the task shaped our approach. First, macro-F1 weights every class equally. The AGAINST class makes up only 15 % of the permitted English pool, yet it carries a full third of the score: a system that never predicts AGAINST confidently forfeits that class’s F1 entirely, and even perfect FAVOUR and NONE predictions then cap macro-F1 at 0.67. Second, and less obviously, the evaluation data is not what the task description leads one to expect. We therefore begin with a diagnostic rather than a model (Section 4).

Our novel contribution is a diagnose-first transfer strategy. We document linguistic and topical evidence that the nominally Hindi and Bengali evaluation set was machine-translated from English YouTube comments, then use this diagnosis to motivate back-translation and source-language classification. This English pivot improves macro-F1 by about 0.10 over the best multilingual-encoder system (XLM-R-large, Table 4), averaged over 300 paired half-splits of the held-out judge-panel development



Table 1

Measurements over all 1 000 evaluation comments.

Signal	Hindi	Bengali
Comments naming one science communicator	63	96
Roman-script code-mixing	0	0
Comments containing any emoji	0	0
Comments mentioning India	2	3
Median length (characters)	146	147

set. The taxonomy, synthetic augmentation, panel distillation, and calibration support this strategy rather than constituting separate novelty claims.

2. Related Work

Stance detection was established as a benchmark task by SemEval-2016 Task 6 [1] and extended to unseen targets by zero-shot formulations [2]. For climate discourse specifically, GWSD provides English news sentences with crowd stance annotations [3]. The CARDS scheme [4] organises contrarian climate claims into a hierarchy of super-claims and sub-claims; our taxonomy adopts it as one branch.

Cross-lingual transfer for Indian languages typically relies on multilingual encoders such as XLM-R [5], MuRIL [6] and IndicBERTv2 [7]. An alternative reformulates classification as entailment against a hypothesis [8], and XNLI-trained models [9] make this available multilingually; we include it as a baseline. Language models are increasingly applied to stance detection directly [10]. We use them as teachers instead, following knowledge distillation [11]: the panel labels unlabelled target data, and a compact encoder learns from those labels, so the submitted run is a local classifier rather than a live API call. Our projection channel adapts a hybrid framework for multilingual zero-shot classification [12]; where that work aligns sentence embeddings with ConceptNet [13] label embeddings, we substitute the task-specific taxonomy, and we retain its Kolmogorov–Arnold alignment [14].

3. Task and Data

We first describe the data as released by the organisers, because every later design decision depends on it. The evaluation data comprises two files of 500 comments each, for Hindi and Bengali, with columns ID and COMMENT and identifiers 1–500 in both files. Runs are submitted as one file per language carrying up to three classifier columns.

The permitted training resources are two English stance corpora: GWSD [3], 2 300 news sentences each annotated by eight crowd workers, and SemEval-2016 Task 6 [1] restricted to the target “*Climate Change is a Real Concern*” (395 tweets). Other public data, machine translation and synthetic generation are explicitly permitted.

Merging and de-duplicating the two corpora yields 2 442 rows; Figure 1 shows their class composition. Two details matter for how we train. For GWSD we retain the full crowd-vote distribution rather than only the majority label: an 8–0 vote and a 5–3 vote are different training signals, and the soft target regularises the small AGAINST class at no cost. On average, 72 % of a GWSD sentence’s eight crowd workers chose its majority label. The pool is also strongly skewed for a macro-F1 task: within the SemEval subset only 15 tweets of 395 are AGAINST.

4. Characterising the Evaluation Set

Before training any model we measured the released files (Table 1). Organic Indian YouTube comments are short, emoji-dense and substantially Roman-script. These comments are none of those things.

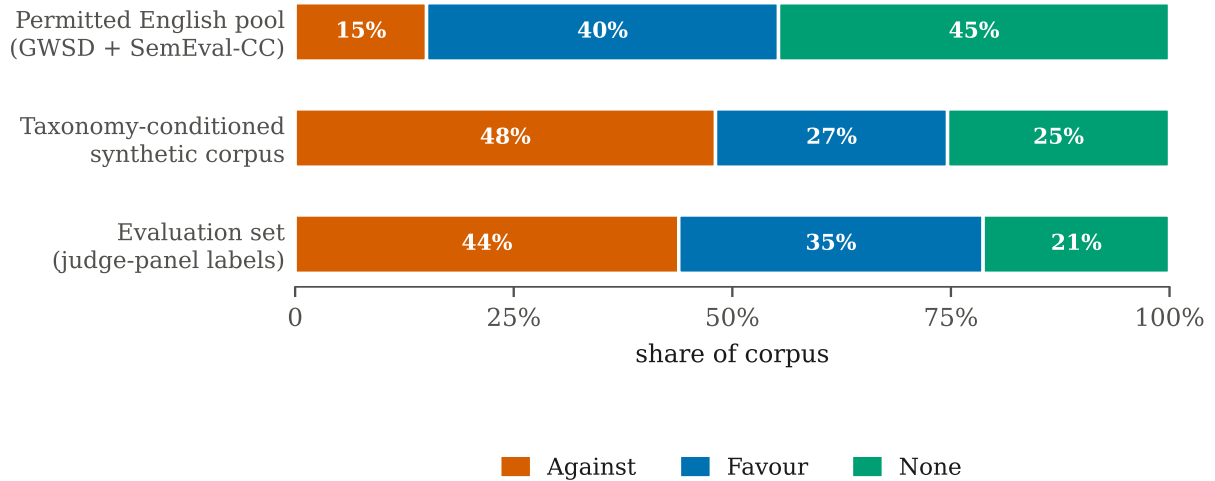


Figure 1: Class composition of the permitted training pool, the synthetic corpus, and the evaluation set as labelled by the judge panel.

Table 2

Back-translated evaluation comments with the taxonomy node (Section 5) and the label assigned to them.

English back-translation	Taxonomy node	Label
“Bill Nye is a person who makes millions by scaring people who don’t think for themselves.”	attacks on the messenger	AGAINST
“Climate change is natural, it’s been happening for 4 billion years.”	not human-caused	AGAINST
“This problem can be solved if all countries around the world cooperate and take environmentally positive actions.”	demands for action	FAVOUR
“I have to watch this in 3rd semester because it’s strike day in the UK.”	class assignment	NONE

The register is uniformly formal. The arguments come from a United States-centric repertoire: attacks on a presenter’s credentials (“*not a scientist, a mechanical engineer*”), disputes over the “97 %” consensus figure, channel-merchandise spam, creationist framing (“*why did God kill the dinosaurs*”), and comments indicating a school assignment (“*who else is here for online classes*”). Numerals appear in Devanagari and Bengali digits, and English names are transliterated. Taken together, these indicators are consistent with English YouTube comments machine-translated into Hindi and Bengali; we treat this as an evidence-based characterisation, not as a claim of recovered provenance. Table 2 shows four back-translated examples.

The two language files are different comment samples, not translations of the same 500 English comments. Matching the multiset of numerals per comment, normalised across Devanagari, Bengali and ASCII digits, recovers no cross-file alignment, so no alignment shortcut is available.

This diagnosis guides the system design in four ways. First, translate-test becomes the primary channel rather than a fallback: back-translation recovers English text in the domain of GWSD, SemEval, and strong English encoders (Section 6.4). Second, telling the annotation models in one prompt sentence that the comments are machine-translated from English YouTube comments under climate-science videos resolves whole families of otherwise ambiguous items (channel praise, presenter attacks, assignment chatter). Third, synthetic data should imitate translationese—the literal, formal register of machine-translated text—rather than authentic Indian slang. Fourth, the argument inventory to cover is the source community’s, not the target language’s. Our India-specific taxonomy branch was designed before this measurement; it accordingly became a minor slice, and we report it as an ablation rather than as the centrepiece.

5. Argument Taxonomy

Cross-lingual stance transfer can also fail through mismatch in the argument inventory. GWSD and SemEval encode a United States contrarian idiom, so their translations do not necessarily cover the arguments encountered in the target data. The taxonomy therefore represents arguments alongside language.

The taxonomy contains 27 nodes—13 AGAINST, 7 FAVOUR and 7 NONE—with 228 cue phrases across English, Hindi and Bengali. The CARDS branch follows Coan et al. [4]: not happening; not human-caused; impacts not harmful; solutions will not work; science unreliable; and *attacks on the messenger*, which we added after the measurements of Section 4 showed it to be among the most frequent arguments actually present. The India-specific contrarian branch covers climate colonialism and whataboutism, development-first arguments, local-weather naturalism, cyclical or religious cosmology, media conspiracy, elite hypocrisy and population-rather-than-climate. The pro-climate branch is grounded in lived experience: floods and air quality, agricultural harm, health, intergenerational duty, demands for action, support for mitigation and trust in science. The NONE inventory covers channel praise, pure questions, neutral factual statements, unrelated politics, spam, ambivalence and—also added after the measurements—comments indicating a class assignment.

Two annotation traps are encoded explicitly, because they are where systems and annotators most often diverge. Sentiment is not stance: “*Shame on this government, they have destroyed our rivers!*” is FAVOUR despite maximally negative sentiment. Praise of the video is not agreement with the claim: “*very informative, thank you*” is NONE even under a climate-alarm video.

The taxonomy is used in three roles: as a generative schema (Section 6.1), as auxiliary supervision (Section 6.3), and as the reasoning scaffold in the annotation prompt (Section 6.2). That triple use is what makes it a resource rather than a lexicon appendix.

6. Method

Figure 2 shows the full pipeline. The characterisation of Section 4 sits at its head: every later component exists because the evaluation data turned out to be translated English. This section describes the components in the order in which data flows through them.

6.1. Taxonomy-conditioned synthetic augmentation

We walk the product node $\times$ language $\times$ register and prompt a language model for realistic comments in each cell. The stance of a comment is fixed by its cell, so the corpus is class-balanced by construction. Following Section 4 we generate in translationese style. The result is 1 040 comments covering all 27 nodes. Provenance columns (node, branch, language, register, generator) are retained for ablation, and near-duplicates are removed by token Jaccard similarity within each (node, language) cell.

Figure 1 provides a descriptive distributional check. We built the synthetic corpus to be AGAINST-heavy from the forensic reading before producing any panel labels. The judge-panel labels on the evaluation set later showed an AGAINST share only four percentage points away from the synthetic corpus’s. This is not an independent evaluation, because the comparison uses model-generated labels, but it supports the motivation for the synthetic class balance.

6.2. Transductive panel distillation

We use the entire *unlabelled* evaluation set for target-adaptive pseudo-labelling; no organiser-provided label is accessed. Two disjoint panels label all 1 000 comments; Table 3 summarises their reliability.

A teacher panel of five inexpensive models from five different laboratories labels every comment. Each model must name a taxonomy node *before* emitting a stance, which constrains its reasoning to a fixed, auditable vocabulary. A judge panel of three stronger models then adjudicates, reading the Indic original together with its English back-translation, while a fourth model adversarially challenges every

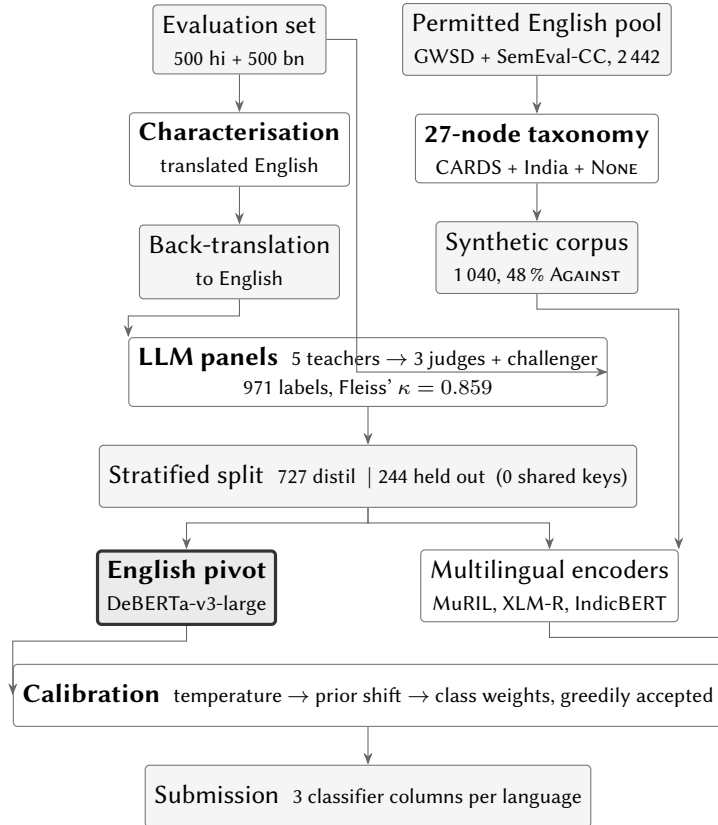


Figure 2: The pipeline. Characterising the evaluation set (upper left) is what makes the English pivot available.

Table 3

Panel reliability. Teacher-panel κ is reported per language (Hindi / Bengali).

Measure	Value
Teacher panel Fleiss' κ (5 models)	0.767 / 0.749
Judge panel Fleiss' κ (3 models)	0.859
Judge labels accepted (of 1 000)	971
Adversarial overturn rate	2.8 %
Teacher-judge agreement (971 comments)	92.7 %
Teacher macro-F1 against judge labels (971 comments)	0.917

proposed label. The two panels are disjoint by construction, and the code refuses to run if any model identifier appears in both roles.

The accepted labels are split, stratified by language and label, into 727 distillation rows and a 244-row held-out development set, with a programmatic check that the two share no (language, identifier) key. Every result in Section 8 is measured against this held-out judge-panel split, not against organiser-provided gold labels.

6.3. Argument-aware encoder

Each encoder carries two heads over a shared representation: a three-way stance head, trained with class-weighted cross-entropy plus a soft-label KL term, and an auxiliary argument-node head of weight $\lambda = 0.3$. The auxiliary signal is free, because synthetic rows are node-labelled by construction and the panels predict a node for every real comment. It pushes the shared representation towards argument structure rather than surface lexical cues, which is the axis along which we need the model to generalise. Rows without a node are masked out of the auxiliary loss. We use class weighting rather

than oversampling, because duplicating rows distorts soft targets whereas reweighting does not.

6.4. The English pivot

The largest gain comes from the component that follows most directly from Section 4. We back-translate all 1 000 evaluation comments into English and fine-tune DeBERTa-v3-large [15] on an English-only pool: the 2 442 GWSD and SemEval rows plus the back-translated judge-labelled rows. Instead of asking a multilingual encoder to bridge English training data and Indic evaluation data, we move the evaluation data to where the training data already lives.

6.5. Macro-F1-aware calibration

Taking $\arg \max_y p(y \mid x)$ maximises accuracy, not macro-F1. We therefore apply three post-hoc stages in order: temperature scaling [16], prior-shift correction by the Saerens–Latinne–Decaestecker expectation-maximisation procedure [17], and a coordinate search over per-class multiplicative weights. Each stage is kept only if it does not lower development macro-F1. The greedy acceptance matters: on Bengali, prior correction would have reduced the distilled MuRIL-base channel from 0.781 to 0.200, and it was rejected automatically. We also floor and damp the expectation-maximisation update, because on an under-confident model the textbook procedure drives $p(\text{AGAINST})$ to zero and removes the class entirely.

7. Experimental Setup

This section collects the details needed to reproduce the runs. We evaluate MuRIL base and large [6], XLM-R base and large [5], IndicBERTv2 [7], DeBERTa-v3-large [15], and LaBSE [18] as a frozen encoder. The maximum sequence length is 128 and the random seed is fixed throughout. Base models were trained on 32 CPU cores and large models on a single rented RTX 4090 in bfloat16, at a total rental cost of about one US dollar. All submitted runs are local checkpoints and require no API call at inference.

We also evaluate a projection channel that adapts the hybrid framework of Deb et al. [12]. LaBSE embeddings are projected into a semantic space built from the taxonomy’s gloss-and-cue embeddings by a hybrid of a Kolmogorov–Arnold network [14] and support-vector regression, and stance is read from cosine similarity aggregated over nodes. The taxonomy here replaces ConceptNet [13] as the knowledge source: it is domain-matched and much richer than a bare three-way label space. Finally we include a training-free entailment baseline [8] using a multilingual XNLI model [9].

8. Results

8.1. Development results

Table 4 reports macro-F1 against the held-out judge-panel development split, which shares no row with any training data. The English pivot has the highest score among the evaluated configurations in both languages. Across 300 paired resampled half-splits of that development set, it improves on the strongest pre-pivot systems (XLM-R-large and the searched fusion of the remaining channels) in every split, with a mean gain of about 0.10 macro-F1.

We verified separation between the English distillation pool and the development split: they share no (language, identifier) keys, no development comment text appears verbatim in the training data, and the GWSD and SemEval pool contains no evaluation comments.

Calibration adds about 0.02 macro-F1 on the pivot channel and up to 0.08 on the weaker base encoders, measured on the same development split. The larger multilingual encoders beat their distilled base counterparts in both languages, and the auxiliary node head and the soft-label term are both retained in the submitted configuration.

Table 4

Macro-F1 against held-out judge-panel labels (123 Hindi, 121 Bengali comments). Submitted runs are in bold; per-class F1 is for the primary run.

System	hi	bn
<i>English pivot</i>		
DeBERTa-v3-L + calibration	0.922	0.915
without calibration	0.904	0.892
DeBERTa-v3-L, English-only pool	0.886	0.877
<i>Native-script encoders</i>		
XLM-R-large	0.836	0.812
MuRIL-large	0.807	0.790
IndicBERTv2	0.804	0.791
MuRIL-base, distilled	0.758	0.781
XLM-R-base, distilled	0.717	0.707
<i>Other channels</i>		
Argument-KG projection	0.763	0.722
Claim-conditioned entailment	0.438	0.404
<i>Per-class F1, submitted primary run</i>		
FAVOUR	0.935	0.921
AGAINST	0.963	0.961
NONE	0.870	0.863

Table 5

Official macro-F1 on the evaluation set (500 comments per language), as returned by the organisers.

Run	System	hi	bn
mode11	DeBERTa-v3-L, English pivot, calibrated	[FILL]	[FILL]
mode12	DeBERTa-v3-L, English-only pool	[FILL]	[FILL]
mode13	XLM-R-large, native script	[FILL]	[FILL]

Both submitted language files carry the calibrated English pivot as the primary classifier, the English-pool DeBERTa variant as the second, and XLM-R-large on native script as the third. Over the 500 submitted comments per language, all three columns give the same label for 87.2 % of Hindi and 86.4 % of Bengali comments.

8.2. Results on the evaluation data

Table 5 reports the macro-F1 returned by the organisers for our three runs on the 500 Hindi and 500 Bengali evaluation comments, the figures on which the task is actually ranked. Our overall standing was [FILL]. Read against Table 4, these numbers are the external check that the development figures could not provide: the development labels come from our own judge panel, so the gap between the two columns measures how far that panel’s decision boundary sits from the organisers’ annotation guidelines rather than any change in the classifiers themselves.

Independently of the official scores, the predictions themselves are informative, and Table 6 gives the label distribution each run assigned to the evaluation set. All three runs place AGAINST first and NONE last in both languages, which corroborates from the model side the class balance we inferred when characterising the evaluation files in Section 4: the evaluation set is far richer in AGAINST than the permitted English training pool, whose AGAINST share is 15 %. The three runs also disagree in a structured way. Moving from the pivot to the native-script encoder moves mass out of FAVOUR, which falls by 3.8 points in Hindi and 6.6 in Bengali, and into NONE, which rises by 3.4 and 3.6; this is the same FAVOUR/NONE confusion that dominates our development error analysis.

Table 6

Percentage of the 500 evaluation comments per language assigned to each class by each submitted run (F: FAVOUR, A: AGAINST, N: NONE).

Run	Hindi			Bengali		
	F	A	N	F	A	N
model11	39.0	44.0	17.0	38.8	40.2	21.0
model12	38.6	42.4	19.0	35.4	42.4	22.2
model13	35.2	44.4	20.4	32.2	43.2	24.6

9. Error Analysis

The submitted system’s weakest class is NONE in both languages on the panel-labelled development split (Table 4). This matches where the taxonomy predicted difficulty. NONE is not a semantic category but the absence of a stance: it collects channel praise, questions, neutral facts and ambivalence, whose boundary with FAVOUR rests on a codebook decision rather than on the text. The most frequent confusions are NONE predicted as FAVOUR, where polite engagement is read as agreement, and AGAINST predicted as FAVOUR, where sarcasm survives translation poorly.

Across all 1 000 comments, the most frequent arguments assigned by the panels are demands for action and science-unreliable in both languages; channel praise ranks third in Bengali and fourth in Hindi, and attacks on the messenger rank fourth in Bengali, which supports the decision to add that node after the measurements of Section 4.

10. Limitations

The most important caveat is that our development labels are not human gold. The 244 development labels come from the judge panel, so every figure in Table 4 measures agreement with that panel rather than with ground truth. The panel’s internal agreement is high (Table 3), and a human reader reviewed and endorsed a fifteen-comment sample of the development set, but that review was not blind and yields no independent agreement statistic. If the panel’s reading of the NONE boundary differs systematically from the organisers’, our official score will be lower than reported here, and our calibration, fitted to the panel’s class priors, may transfer imperfectly. We regard a blind human annotation of 50–150 comments as the single highest-value extension of this work.

The pivot also requires machine translation at inference. The classifier itself is a local checkpoint with no API dependency, but reproducing the run requires either the back-translated files, which we release with this paper, or a re-translation step. Machine translation is a standard and organiser-encouraged step for this task, but it remains a pipeline dependency. Finally, the synthetic corpus reflects a language model’s *model* of Indian climate discourse rather than the discourse itself, and the development set is small, at roughly 120 rows per language, so ensemble-level comparisons would be unreliable at this size.

11. Conclusion

The lesson we would carry to another cross-lingual shared task is procedural: characterise the evaluation distribution before choosing a transfer strategy. Here the nominal problem was Hindi and Bengali social media, while the released comments showed strong evidence of translated English provenance. Recognising that distinction motivated the English-pivot strategy. The argument taxonomy, synthetic corpus, and panel labels are released with this paper so that the components can be reused and examined further.

Declaration on Generative AI

ChatGPT was used to improve grammar, correct spelling mistakes, and rephrase parts of the manuscript. The author reviewed and checked all revised text, verified the reported content and results, and takes full responsibility for this publication.

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